

Chapter 3: Heavy-Tailed Distributions

A technique-oriented study guide for the exercises

Based on M. Bladt, *Statistics for Insurance*, Ch. 3 (Week 4 notes)

How to read this. Section 1 is the one-page mental model. Section 2 is the core: the 12 techniques that solve almost every exercise in the chapter, each with *when to use it* and a worked example. Sections 3–8 go through the chapter’s definitions and theorems in order, saying what each one is *for*. Section 9 maps every exercise in §3.8 to the technique(s) it needs, and Section 10 lists the classic mistakes.

Notation. $\bar{F} = 1 - F$ is the survival function, $\gamma > 0$ the *extreme value index*, and $\alpha = 1/\gamma$ the *tail index*. The notes write the index as $1/\gamma$, so watch out for γ versus $1/\gamma$.

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1 The big picture in one page

The central object. A positive random variable Z is *regularly varying* (RV) with index $1/\gamma$ if

$$\overline{F}(z) = \mathbb{P}(Z > z) = z^{-1/\gamma} \ell(z), \quad \ell \text{ slowly varying: } \frac{\ell(tz)}{\ell(z)} \rightarrow 1 \quad \forall t > 0.$$

Equivalently (this is the form you use most):

$$\lim_{z \rightarrow \infty} \frac{\overline{F}(tz)}{\overline{F}(z)} = t^{-1/\gamma} \quad \text{for all } t > 0.$$

So beyond a high threshold the tail looks like a *Pareto* tail. Everything in the chapter follows from this:

1. **Classification (§3.1).** RV $\subset$ subexponential $\subset$ long-tailed $\subset$ heavy-tailed.
2. **Why RV is natural (§3.2).** Maxima can only converge to Gumbel/Fréchet/Weibull (GEV). The Fréchet domain is *exactly* the RV class (Thm 3.2). Threshold exceedances converge to the GPD (Thm 3.3).
3. **Tools to handle ℓ (§3.2.2).** These are uniform convergence (3.4), Potter bounds (3.5, 3.6), the representation theorem (3.7) and Karamata (3.8). Their job is to control the unknown ℓ so you can take limits and use dominated convergence.
4. **Estimation (§3.3).** Take the k largest observations, with $k \rightarrow \infty$ and $k/n \rightarrow 0$. Rescaled by the threshold, they look like a Pareto sample, and $\log(\text{Pareto})$ is exponential with mean γ . That gives the **Hill estimator**:

$$\gamma_{n,k} = \frac{1}{k} \sum_{i=1}^k \log \frac{Z_{n,n-i+1}}{Z_{n,n-k}}, \quad \sqrt{k}(\gamma_{n,k} - \gamma) \xrightarrow{d} N(0, \gamma^2).$$

5. **Extrapolation (§3.3.3).** The Weissman estimators use $\gamma_{n,k}$ to go beyond the data, giving high quantiles and small tail probabilities.
6. **Other estimators (§3.4).** Pickands, GPD-MLE and the moment estimator work for all $\gamma \in \mathbb{R}$.
7. **Covariates (§3.5–3.6).** The conditional Hill estimator $\gamma_{n,k}(x)$ is a kernel-weighted Hill estimator. Its rate is $\sqrt{k_n h_n}$ instead of $\sqrt{k}$, and its asymptotic variance is $\gamma(x)^2 \int K^2/g(x)$.

The single most important identity (it links probability to statistics):

$$\gamma = \int_1^\infty \frac{T(s)}{s} ds, \quad T(s) = s^{-1/\gamma}, \quad \gamma_{n,k} = \int_1^\infty \frac{\widehat{T}_n(s)}{s} ds.$$

The Hill estimator is the same functional applied to the empirical tail function. Consistency and asymptotic normality are then proved by showing $\widehat{T}_n \rightarrow T$ in a strong enough sense and controlling the integral near ∞ .

2 The toolbox: 12 techniques that solve the exercises

T1. Prove that a distribution is regularly varying

Use either of two routes.

- (a) **Ratio test.** Compute $\lim_{z \rightarrow \infty} \overline{F}(tz)/\overline{F}(z)$ and show it equals $t^{-1/\gamma}$ for every $t > 0$.

(b) **Factor out the power.** Write $\bar{F}(z) = z^{-1/\gamma}\ell(z)$ and show ℓ is slowly varying. The quickest sufficient condition is

$$\ell(z) \rightarrow c \in (0, \infty) \implies \ell \text{ is slowly varying.}$$

When to use it

Any “show that X is regularly varying and find the index” question (Ex. 5, Example 3.1). Route (b) is usually fastest when $\bar{F}$ is explicit.

Example: the Burr distribution (Exercise 5)

$\bar{F}(x) = (1 + (x/\theta)^\eta)^{-\beta}$. Factor out the dominant power:

$$\bar{F}(x) = (x/\theta)^{-\eta\beta} (1 + (\theta/x)^\eta)^{-\beta} = x^{-\eta\beta} \underbrace{\theta^{\eta\beta} (1 + (\theta/x)^\eta)^{-\beta}}_{\ell(x) \rightarrow \theta^{\eta\beta} \in (0, \infty)}.$$

So ℓ is slowly varying and Burr is RV with index $1/\gamma = \eta\beta$, i.e. $\gamma = 1/(\eta\beta)$. With the ratio test, $\bar{F}(tx)/\bar{F}(x) = \left(\frac{1+(tx/\theta)^\eta}{1+(x/\theta)^\eta}\right)^{-\beta} \rightarrow (t^\eta)^{-\beta} = t^{-\eta\beta}$.

Further quick examples.

- Pareto: $\ell \equiv 1$.
- Shifted Pareto: $(x+1)^{-1/\gamma} = x^{-1/\gamma}(1+1/x)^{-1/\gamma}$.
- Fréchet: $1 - e^{-x^{-1/\gamma}} = x^{-1/\gamma} \cdot x^{1/\gamma}(1 - e^{-x^{-1/\gamma}})$, and the second factor tends to 1 since $1 - e^{-y} \sim y$ as $y \downarrow 0$.

T2. From density to tail: Karamata / L'Hôpital

If only the density is nice, show $f(x) = x^{-1/\gamma-1}\ell(x)$ with ℓ slowly varying. Karamata (Thm 3.8) then gives

$$\bar{F}(x) = \int_x^\infty f(t) dt \sim \gamma x^{-1/\gamma} \ell(x).$$

In particular, if $f(x) \sim c x^{-\alpha-1}$, then $\bar{F}(x) \sim \frac{c}{\alpha} x^{-\alpha}$. In the ratio test you can also use L'Hôpital directly:

$$\lim \frac{\bar{F}(tx)}{\bar{F}(x)} = \lim \frac{tf(tx)}{f(x)}.$$

When to use it

The distribution is given by its density and $\bar{F}$ has no closed form, e.g. Student- t , Gamma-type, or “show that a density with power decay gives an RV tail”.

Example: Student t_ν

$f_\nu(x) = c_\nu(1 + x^2/\nu)^{-(\nu+1)/2} = x^{-(\nu+1)}\ell(x)$ with $\ell(x) = c_\nu x^{\nu+1}(1 + x^2/\nu)^{-(\nu+1)/2} \rightarrow c_\nu \nu^{(\nu+1)/2}$. Hence $\bar{F}_\nu(x) \sim \frac{1}{\nu} x^{-\nu}\ell(x)$ and the tail is RV with index ν , so $\gamma = 1/\nu$.

The degrees of freedom *are* the tail index. Cauchy ($\nu = 1$) has $\gamma = 1$.

T3. Prove that a distribution is *not* regularly varying

Show that $\bar{F}(tx)/\bar{F}(x)$ does not converge to a power of t . Typically you show

$$\frac{\bar{F}(tx)}{\bar{F}(x)} \rightarrow 0 \quad \text{for some (all) } t > 1,$$

which means the tail is *lighter* than every power. Alternatively, show the limit does not exist.

When to use it

Weibull, lognormal, exponential, normal (Ex. 4(b), Example 3.2), and the counterexamples in Ex. 2.

Examples

- **Weibull** $\bar{F}(x) = e^{-x^a}$: $\frac{\bar{F}(tx)}{\bar{F}(x)} = \exp(-x^a(t^a - 1)) \rightarrow 0$ for $t > 1$, for *every* $a > 0$.
- **Lognormal**: by Mills' ratio, $\bar{F}(x) \sim \frac{\sigma}{\log x \sqrt{2\pi}} e^{-(\log x)^2/(2\sigma^2)}$. Since $(\log tx)^2 - (\log x)^2 = 2 \log t \log x + (\log t)^2 \rightarrow \infty$, the ratio tends to 0.

So lognormal and Weibull with $a < 1$ are *heavy-tailed but not RV*. They lie in the *Gumbel* domain of attraction.

T4. Prove that a distribution is heavy-tailed (or not)

Heavy-tailed means $M_Z(t) = \mathbb{E}[e^{tZ}] = \infty$ for all $t > 0$. For $Z \geq 0$ use

$$\mathbb{E}[e^{tZ}] = 1 + t \int_0^\infty e^{tz} \bar{F}(z) dz.$$

This is integration by parts, or Fubini on $e^{tZ} - 1 = \int_0^Z te^{tz} dz$. So it suffices to show $e^{tz} \bar{F}(z) \rightarrow \infty$ (criterion (3.1)), or even just that the integrand does not go to 0.

When to use it

Ex. 4(a), Example 3.2, and showing that “long-tailed $\Rightarrow$ heavy-tailed” or “RV $\Rightarrow$ heavy-tailed”.

Examples

Weibull (Ex. 4a). $\mathbb{E}e^{tZ} = 1 + t \int_0^\infty e^{tz-z^a} dz$.

- If $a < 1$: $tz - z^a \rightarrow \infty$, so the integral is infinite for every $t > 0$ and Weibull is heavy-tailed.
- If $a = 1$: the integral is finite for $t < 1$.
- If $a > 1$: the integral is finite for all t .

So Weibull is heavy-tailed $\iff a < 1$.

RV $\Rightarrow$ heavy. By Potter's lower bound (Thm 3.5), $\bar{F}(z) \geq c z^{-1/\gamma-\varepsilon}$, so $e^{tz} \bar{F}(z) \geq c e^{tz} z^{-1/\gamma-\varepsilon} \rightarrow \infty$.

T5. Potter bounds $\Rightarrow$ a dominating function $\Rightarrow$ dominated convergence

The tail ratio $\bar{F}(x/y)/\bar{F}(x) \rightarrow y^{1/\gamma}$ holds pointwise, but you usually need it *inside an integral*. Potter's second version (Thm 3.6) gives a bound uniform in $x \geq 1$:

$$\frac{\bar{F}(x/y)}{\bar{F}(x)} \leq C(y^{1/\gamma-\varepsilon} \vee y^{1/\gamma+\varepsilon}).$$

Equivalently, $\frac{\bar{F}(ut)}{\bar{F}(u)} \leq C t^{-1/\gamma+\varepsilon}$ for $t \geq 1$. This bound is integrable against dt/t on $[m, \infty)$, and the integral is $O(m^{-1/\gamma+\varepsilon})$.

When to use it

Whenever you must pass a limit through $\int$ or $\mathbb{E}$ involving a tail ratio:

- Breiman's lemma (Ex. 7);
- the “tail piece” $\int_m^\infty$ in the Hill/moment/conditional Hill proofs;
- variance bounds such as $\mathbb{E}[\log_+^2(Z/(mu_n))]/\bar{F}(u_n)$;
- Ex. 6.

Worked example: Breiman's lemma (Exercise 7)

Let $Z \perp W$, Z RV with index $1/\gamma$, $W \geq 0$ and $\mathbb{E}W^{1/\gamma+\varepsilon} < \infty$. Condition on W :

$$\frac{\mathbb{P}(ZW > x)}{\mathbb{P}(Z > x)} = \int_{(0, \infty)} \frac{\bar{F}_Z(x/w)}{\bar{F}_Z(x)} \mathbb{P}_W(dw).$$

The atom at $w = 0$ contributes nothing.

1. *Pointwise*: $\bar{F}_Z(x/w)/\bar{F}_Z(x) \rightarrow w^{1/\gamma}$ (RV with $t = 1/w$).
2. *Domination* (Potter v2, $x \geq 1$): the integrand is $\leq C(w^{1/\gamma-\varepsilon} \vee w^{1/\gamma+\varepsilon}) \leq C(1 + w^{1/\gamma+\varepsilon})$, which is $\mathbb{P}_W$ -integrable.
3. *DCT*: the limit is $\mathbb{E}[W^{1/\gamma}]$.

RV of ZW . Assume $\mathbb{P}(W > 0) > 0$, so that $\mathbb{E}W^{1/\gamma} > 0$. Then

$$\frac{\mathbb{P}(ZW > tx)}{\mathbb{P}(ZW > x)} = \underbrace{\frac{\mathbb{P}(ZW > tx)}{\mathbb{P}(Z > tx)}}_{\rightarrow \mathbb{E}W^{1/\gamma}} \cdot \underbrace{\frac{\mathbb{P}(Z > tx)}{\mathbb{P}(Z > x)}}_{\rightarrow t^{-1/\gamma}} \cdot \underbrace{\frac{\mathbb{P}(Z > x)}{\mathbb{P}(ZW > x)}}_{\rightarrow 1/\mathbb{E}W^{1/\gamma}} \rightarrow t^{-1/\gamma}.$$

Moral: multiplying by an independent factor with lighter tails keeps the tail index and only scales the constant. Example: claim = (RV loss) $\times$ (bounded inflation factor).

T6. Uniform convergence of tail ratios (Thm 3.4) and plugging in random levels

$$\sup_{t \geq a} \left| \frac{\bar{F}(tz)}{\bar{F}(z)} - t^{-1/\gamma} \right| \rightarrow 0 \quad (a > 0 \text{ fixed}).$$

Because the convergence is uniform, you may substitute a *random* or n -dependent argument t_n such as $sZ_{n,n-k}/u_n$:

$$\frac{\bar{F}(u_n t_n)}{\bar{F}(u_n)} - t_n^{-1/\gamma} \rightarrow 0.$$

When to use it

Replacing the unobservable threshold $u_n = F^{\leftarrow}(1 - k/n)$ by $Z_{n,n-k}$ (Thm 3.12, Hill CLT), or when an argument moves with n . It is also the reason that the bias term $B_n(s_0) = \sup_{t \geq s_0} |T_n(t) - T(t)| \rightarrow 0$.

T7. Bias + stochastic decomposition, and Chebyshev

For an estimator $\tilde{T}_n$ of a deterministic target T , write

$$\tilde{T}_n - T = \underbrace{(\tilde{T}_n - \mathbb{E}\tilde{T}_n)}_{\text{stochastic, mean 0}} + \underbrace{(\mathbb{E}\tilde{T}_n - T)}_{\text{deterministic bias}}.$$

Show that the variance of the first term goes to 0 (Chebyshev gives $\xrightarrow{\mathbb{P}} 0$) and that the bias goes to 0 (by RV). Dini's theorem (Thm 1.15) then upgrades pointwise to *uniform* convergence, because $s \mapsto \tilde{T}_n(s)$ is monotone.

When to use it

Consistency of $\tilde{T}_n(s)$ (Thm 3.9) and of $\tilde{T}_n^x(s)$ (Thm 3.20). It is the first step of every proof in §3.3 and §3.6.

Example: the computation in Thm 3.9

$\tilde{T}_n(s) = \frac{1}{k} \sum_i \mathbf{1}(Z_i > u_n s)$ is a sum of i.i.d. Bernoullis with $p = \bar{F}(u_n s)$, and $n\bar{F}(u_n) = k$. Then

$$\text{Var } \tilde{T}_n(s) = \frac{n p(1-p)}{k^2} \leq \frac{n \bar{F}(u_n s)}{k^2} = \frac{1}{k} \cdot \frac{\bar{F}(u_n s)}{\bar{F}(u_n)} \approx \frac{s^{-1/\gamma}}{k} \rightarrow 0.$$

Moral: the effective sample size is k , not n . In the conditional case it is $k_n h_n$.

T8. Order statistics $\leftrightarrow$ tail empirical function (the “inversion trick”)

Order-statistic events can be rewritten as counting events:

$$\{Z_{n,n-k} > u_n(1 + \varepsilon)\} \subseteq \{\text{at least } k \text{ obs. exceed } u_n(1 + \varepsilon)\} = \{\tilde{T}_n(1 + \varepsilon) \geq 1\}.$$

Since $\tilde{T}_n(1 + \varepsilon) \xrightarrow{\mathbb{P}} (1 + \varepsilon)^{-1/\gamma} < 1$, this has probability $\rightarrow 0$, and therefore $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$. The same argument gives $q_n^x/u_n^x \xrightarrow{\mathbb{P}} 1$ in the conditional case (Prop. 3.3).

When to use it

Whenever you need the random threshold $Z_{n,n-k}$ (or q_n^x) to behave like the deterministic u_n (or u_n^x).

T9. Sum $\leftrightarrow$ integral representation

Use the elementary identity

$$\int_1^\infty \frac{\mathbf{1}(z > at)}{t} dt = \log_+(z/a).$$

Summing it over the sample turns Hill-type sums into integrals of the tail empirical function, and the reverse. Likewise

$$\int_1^\infty \frac{\log t}{t} \mathbf{1}(z > at) dt = \frac{1}{2} \log_+^2(z/a)$$

handles the second moment $M_n^{(2)}$.

When to use it

Ex. 3, consistency of the moment estimator (Ex. 9), the definition of the conditional Hill estimator, and variance calculations of the form $\text{Var} \int_m^\infty \cdots dt$.

Worked example: Exercise 3

(i) $\int_1^\infty \frac{T(s)}{s} ds = \int_1^\infty s^{-1/\gamma-1} ds = \left[-\gamma s^{-1/\gamma} \right]_1^\infty = \gamma.$

(ii) Let $\hat{T}_n(t) = \frac{1}{k} \sum_{i=1}^n \mathbf{1}(Z_i > Z_{n,n-k}t)$. By Fubini and the identity above,

$$\int_1^\infty \frac{\hat{T}_n(t)}{t} dt = \frac{1}{k} \sum_{i=1}^n \log_+ \frac{Z_i}{Z_{n,n-k}} = \frac{1}{k} \sum_{i=1}^k \log \frac{Z_{n,n-i+1}}{Z_{n,n-k}} = \gamma_{n,k}.$$

Only the top k observations exceed $Z_{n,n-k}$ (assuming no ties). So the Hill estimator is the plug-in estimator of $\gamma = \int_1^\infty T(s)/s ds$.

T10. “Truncate at m ”: weak convergence by approximation (Thm 1.5)

To show $\int_1^\infty X_n(t) dt \rightarrow \int_1^\infty X(t) dt$ when you only control X_n on compacts:

1. On $[1, m]$, use (uniform) convergence on compacts plus continuous mapping, so that $\int_1^m X_n \rightarrow \int_1^m X$.
2. Show $\int_m^\infty X \rightarrow 0$ as $m \rightarrow \infty$.
3. Show the remainder is uniformly small:

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\left| \int_m^\infty X_n \right| > \eta \right) = 0.$$

Use Markov or Chebyshev, and bound the mean or variance with **Potter** (T5), which gives a bound like $O(m^{-1/\gamma+\epsilon})$.

When to use it

Every consistency/CLT proof for Hill-type estimators: Thms 3.13, 3.14, 3.21, 3.26, Prop. 3.2 (Ex. 9) and Ex. 13. In consistency proofs use Markov on $\mathbb{E}$; in CLT proofs use Chebyshev on Var .

T11. δ -method, functional δ -method and Vervaat (inversion)

- **δ -method (Thm 1.10).** If $a_n(T_n - c) \xrightarrow{d} Z$, then $a_n(h(T_n) - h(c)) \xrightarrow{d} h'(c)Z$.
- **Inverse map (Lemma 1.10 / Vervaat).** If $\sqrt{k}(\tilde{T}_n - T) \xrightarrow{d} \mathbb{T}$, then the quantile process satisfies $\sqrt{k}(\tilde{Q}_n - Q) \xrightarrow{d} -Q' \cdot \mathbb{T} \circ Q$. At level 1 this gives

$$\sqrt{k} \left(\frac{Z_{n,n-k}}{u_n} - 1 \right) \xrightarrow{d} \gamma \mathbb{T}(1) \sim N(0, \gamma^2).$$

- **Random level.** $\hat{T}_n(s) = \tilde{T}_n(sZ_{n,n-k}/u_n)$. Expand: the random level contributes $-T(s)\mathbb{T}(1)$, and Brownian motion becomes a Brownian bridge (Thm 3.12).

When to use it

To go from tail-process limits to limits for $Z_{n,n-k}$, q_n^x , Hill and Weissman, and to handle the integral near 1, $\int_{Z_{n,n-k}/u_n}^1 T(t)/t dt$. Its derivative at 1 is $-T(1) = -1$, which gives the term

$$-\gamma \mathbb{W}(1).$$

T12. Variance of Gaussian functionals

Write the limit as a linear functional of a Gaussian process with a known covariance and integrate the covariance kernel. The kernels you need are:

$$\begin{aligned} \mathbb{T} = \mathbb{W} \circ T : \quad & \text{Cov}(\mathbb{T}(s), \mathbb{T}(t)) = T(s \vee t) = (s \vee t)^{-1/\gamma}, \\ \mathbb{B} \text{ Brownian bridge} : \quad & \text{Cov}(\mathbb{B}(s), \mathbb{B}(t)) = s \wedge t - st, \\ \mathbb{V}^x \text{ (conditional)} : \quad & \text{Cov}(\mathbb{V}^x(s), \mathbb{V}^x(t)) = (s \vee t)^{-1/\gamma(x)} g(x) \int K^2. \end{aligned}$$

When to use it

Ex. 8 (Hill) and Ex. 12 (conditional Hill). It also lets you recover any asymptotic variance in the chapter.

Worked example: Exercises 8 and 12 in one computation

Let V be centred Gaussian with $\text{Cov}(V(s), V(t)) = c(s \vee t)^{-\alpha}$, where $\alpha = 1/\gamma$, and let $L = \int_1^\infty \frac{V(s)}{s} ds - \gamma V(1)$. Use $\int_1^\infty t^{-\alpha-1} dt = \gamma$ and $\int_1^\infty t^{-\alpha-1} \log t dt = \gamma^2$:

$$\begin{aligned} \text{Var} \int_1^\infty \frac{V(s)}{s} ds &= c \int_1^\infty \int_1^\infty \frac{(s \vee t)^{-\alpha}}{st} ds dt = 2c \int_1^\infty t^{-\alpha-1} \int_1^t \frac{ds}{s} dt = 2c\gamma^2, \\ \text{Cov} \left(\int_1^\infty \frac{V(s)}{s} ds, V(1) \right) &= c \int_1^\infty \frac{s^{-\alpha}}{s} ds = c\gamma, \quad \text{Var } V(1) = c. \end{aligned}$$

Hence $\text{Var } L = 2c\gamma^2 - 2\gamma \cdot c\gamma + \gamma^2 c = c\gamma^2$.

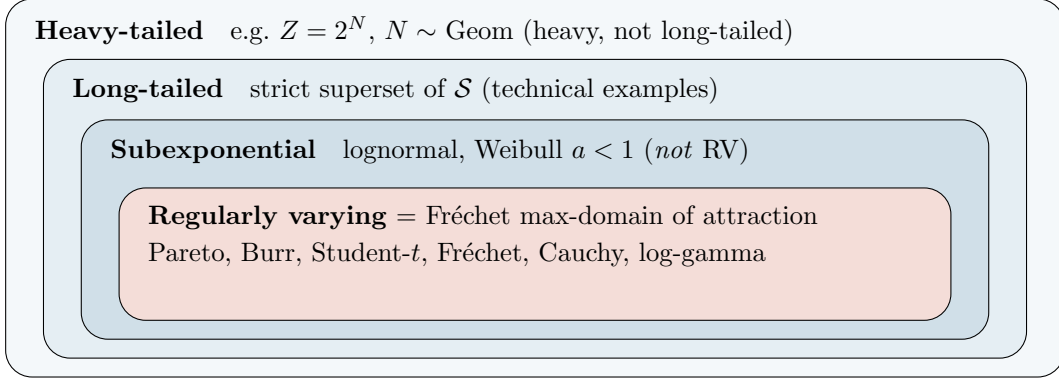
- **Hill (Ex. 8):** $V = \mathbb{W} \circ T$, $c = 1$, so the variance is γ^2 .
- **Conditional Hill (Ex. 12):** the limit is $L/g(x)$ with $c = g(x) \int K^2$, so the variance is $\gamma(x)^2 \int K^2/g(x)$.

Bridge version of Ex. 8. Substituting $s = T(t)$ gives $\int_1^\infty \frac{\mathbb{B}(T(t))}{t} dt = \gamma \int_0^1 \frac{\mathbb{B}(s)}{s} ds$. Its variance is

$$\gamma^2 \int_0^1 \int_0^1 \frac{s \wedge t - st}{st} ds dt = \gamma^2 \left(\int_0^1 \int_0^1 \frac{ds dt}{s \vee t} - 1 \right) = \gamma^2(2 - 1) = \gamma^2.$$

3 §3.1 The heavy-tail classes and how to relate them

Class	Definition	Intuition
Heavy-tailed	$\mathbb{E}e^{tZ} = \infty$ for all $t > 0$; equivalently $e^{tz}\overline{F}(z) \rightarrow \infty$	Tail decays slower than any exponential.
Long-tailed ($\mathcal{L}$)	$\overline{F}(z+x)/\overline{F}(z) \rightarrow 1$ for all x	A big value is not affected by a shift.
Subexponential ($\mathcal{S}$)	$\mathbb{P}(Z_1 + Z_2 > z) \sim 2\overline{F}(z)$	Single big jump: the sum is big only because one term is big.
Regularly varying	$\overline{F}(tz)/\overline{F}(z) \rightarrow t^{-1/\gamma}$	Pareto-like; a scale shift changes the tail by a power.



Exercise 1: the chain of inclusions (proof skeletons)

RV $\Rightarrow$ subexponential (for $Z \geq 0$)

Lower bound: $\{Z_1 + Z_2 > z\} \supseteq \{Z_1 \vee Z_2 > z\}$, so $\mathbb{P}(Z_1 + Z_2 > z) \geq 1 - F(z)^2 = \bar{F}(z)(2 - \bar{F}(z)) \sim 2\bar{F}(z)$.

Upper bound: for $\delta \in (0, \frac{1}{2})$, if $Z_1 + Z_2 > z$ then either one term exceeds $(1 - \delta)z$, or both exceed δz . So

$$\frac{\mathbb{P}(Z_1 + Z_2 > z)}{\bar{F}(z)} \leq 2 \frac{\bar{F}((1 - \delta)z)}{\bar{F}(z)} + \frac{\bar{F}(\delta z)}{\bar{F}(z)} \bar{F}(\delta z) \rightarrow 2(1 - \delta)^{-1/\gamma} + \delta^{-1/\gamma} \cdot 0.$$

Letting $\delta \downarrow 0$ gives $\limsup \leq 2$. **Technique:** split the event, then apply RV ratios (T1).

Subexponential $\Rightarrow$ long-tailed (for $Z \geq 0$)

Condition on Z_2 : $\frac{\mathbb{P}(Z_1 + Z_2 > z)}{\bar{F}(z)} = 1 + \int_{[0, z]} \frac{\bar{F}(z - y)}{\bar{F}(z)} dF(y)$. For fixed $0 < x < z$, split the integral at x and use that $\bar{F}$ is decreasing:

$$\frac{\mathbb{P}(Z_1 + Z_2 > z)}{\bar{F}(z)} \geq 1 + F(x) + \frac{\bar{F}(z - x)}{\bar{F}(z)} (F(z) - F(x)).$$

Rearrange:

$$1 \leq \frac{\bar{F}(z - x)}{\bar{F}(z)} \leq \frac{\mathbb{P}(Z_1 + Z_2 > z)/\bar{F}(z) - 1 - F(x)}{F(z) - F(x)} \rightarrow \frac{2 - 1 - F(x)}{1 - F(x)} = 1.$$

Long-tailed $\Rightarrow$ heavy-tailed

Fix $t > 0$ and let $h(z) = e^{tz}\bar{F}(z)$. Long-tailedness gives $h(z+1)/h(z) = e^t \bar{F}(z+1)/\bar{F}(z) \rightarrow e^t > 1$. So eventually $h(z+1) \geq q h(z)$ with $q = (1 + e^t)/2 > 1$, and h explodes geometrically along $z_0 + n$. For general $z \in [z_0 + n, z_0 + n + 1]$, monotonicity of $\bar{F}$ gives $h(z) \geq e^{-t} h(z_0 + n + 1) \rightarrow \infty$. Now apply criterion (3.1).

Exercise 2: counterexamples for the reverse inclusions

- **Subexponential but not RV:** the lognormal, or Weibull with $a < 1$. These are subexponential by a classical result (Embrechts et al. 1997) and not RV by T3.
- **Heavy but not long-tailed:** $Z = 2^N$ with $\mathbb{P}(N \geq n) = 2^{-n}$, $n \geq 0$.

– Heavy: $\mathbb{E}e^{tZ} \geq e^{t2^n} 2^{-n-1} \rightarrow \infty$.

- Not long-tailed: take $z_n = 2^n - \frac{1}{2}$ and $x = 1$. Then $\bar{F}(z_n + 1)/\bar{F}(z_n) = \mathbb{P}(N \geq n + 1)/\mathbb{P}(N \geq n) = \frac{1}{2} \not\rightarrow 1$.

The **technique** is to look for a distribution whose mass sits on sparse jumps; then a shift can jump over an atom.

- **Long-tailed but not subexponential:** such distributions exist, but the constructions are technical. See Foss, Korshunov & Zachary, *An Introduction to Heavy-Tailed and Subexponential Distributions*. You are unlikely to need to build one from scratch, but you should know that the inclusion is strict.

4 §3.2 Extreme value theory: what each theorem is for

Result	Statement (short)	Use it to...
Fisher–Tippett (3.1)	$\frac{M_n - b_n}{a_n} \xrightarrow{d} \text{GEV}_\gamma$	identify the only possible limits of maxima; block-maxima modelling.
Thm 3.2	Fréchet MDA $\iff \bar{F}$ RV with index $1/\gamma$	move between “maxima” and “tail” language; justify focusing on RV.
GPD (Def 3.1)	$G(y) = 1 - (1 + \gamma \frac{y - \mu}{\sigma})^{-1/\gamma}$	model exceedances; POT-MLE (§3.4).
Pickands–Balkema–de Haan (3.3)	excesses over u , scaled by $a(u)$, $\rightarrow$ GPD	justify peaks-over-threshold; same γ as the GEV.

Example: computing max-limits by hand (technique: $(1 - x_n/n)^n \rightarrow e^{-x}$)

Use $\mathbb{P}(M_n \leq a_n z + b_n) = F(a_n z + b_n)^n$ and choose a_n, b_n so that $n\bar{F}(a_n z + b_n) \rightarrow \tau(z)$. Then the limit is $e^{-\tau(z)}$.

- **Pareto** $\bar{F}(z) = z^{-1/\gamma}$, with $a_n = n^\gamma$, $b_n = 0$: $n\bar{F}(n^\gamma z) = z^{-1/\gamma}$, so the limit is $\exp(-z^{-1/\gamma})$, Fréchet ($\gamma > 0$).
- **Exponential**, with $a_n = 1$, $b_n = \log n$: $ne^{-(z + \log n)} = e^{-z}$, so the limit is $\exp(-e^{-z})$, Gumbel ($\gamma = 0$).
- **Uniform(0,1)**, with $a_n = 1/n$, $b_n = 1$, $z < 0$: $n \cdot (-z/n) = -z$, so the limit is e^z , reversed Weibull ($\gamma = -1$).

For a general RV F take $a_n = F^{\leftarrow}(1 - 1/n)$. Then $n\bar{F}(a_n z) = \bar{F}(a_n z)/\bar{F}(a_n) \rightarrow z^{-1/\gamma}$; this is the “if” direction of Thm 3.2 in one line.

Example: Pickands–Balkema–de Haan for Pareto

$$\mathbb{P}\left(\frac{Z - u}{\gamma u} > y \mid Z > u\right) = \frac{(u(1 + \gamma y))^{-1/\gamma}}{u^{-1/\gamma}} = (1 + \gamma y)^{-1/\gamma}.$$

This is *exactly* the standard GPD with $a(u) = \gamma u$. For a general RV distribution it holds asymptotically, by T6.

Domains of attraction to remember:

- *Weibull MDA* ($\gamma < 0$): finite right endpoint, so not heavy-tailed. Examples: uniform, beta.
- *Gumbel MDA* ($\gamma = 0$): exponential, normal, gamma, but *also* the heavy-tailed lognormal and Weibull with $a < 1$.
- *Fréchet MDA* ($\gamma > 0$): exactly the RV distributions.

5 §3.2.2 Properties of regularly varying functions

Result	Statement	When to use
Uniform convergence (3.4)	$\sup_{t \geq a} \overline{F}(tz)/\overline{F}(z) - t^{-1/\gamma} \rightarrow 0$	Plug in n -dependent or random t (T6); show $B_n(s_0) \rightarrow 0$; make brackets independent of n in bracketing CLTs.
Potter v1 (3.5)	$cz^{-1/\gamma-\varepsilon} \leq \overline{F}(z) \leq Cz^{-1/\gamma+\varepsilon}$	Show RV $\Rightarrow$ heavy-tailed; moment existence ($\mathbb{E}Z^p < \infty$ iff $p < 1/\gamma$, roughly).
Potter v2 (3.6)	$\frac{\overline{F}(x/y)}{\overline{F}(x)} \leq C(y^{1/\gamma-\varepsilon} \vee y^{1/\gamma+\varepsilon})$	Dominated convergence (T5); Breiman, tail integrals $\int_m^\infty$, variance bounds.
Representation (3.7)	$\overline{F}(z) = c(z) \exp \int_{z_0}^z \frac{\delta(u)}{u} du,$ $c \rightarrow c_0, \delta \rightarrow -1/\gamma$	Prove RV of parametric families; derive properties of ℓ (e.g. $z^{-\varepsilon} \ell(z) \rightarrow 0$).
Karamata (3.8)	f RV with index $-1/\gamma - 1$ $\Rightarrow \int_z^\infty f \sim \gamma z f(z)$	Density $\rightarrow$ tail (T2); integrals of RV functions; mean excess asymptotics.

Exercise 6: proving Potter v1 (the strategy in words)

Let $\ell(z) = z^{1/\gamma} \overline{F}(z)$, which is slowly varying.

- *Upper bound.* $z \mapsto z^{-\varepsilon} \ell(z)$ is RV with negative index, so it tends to 0. Hence it is bounded on $[b', \infty)$. It is also bounded on the compact $[1, b']$ (it is a product of bounded functions), and ≤ 1 on $(0, 1)$. So $\sup_{z > 0} z^{-\varepsilon} \ell(z) =: C < \infty$, and $\overline{F}(z) = z^{-1/\gamma+\varepsilon} \cdot z^{-\varepsilon} \ell(z) \leq Cz^{-1/\gamma+\varepsilon}$.
- *Lower bound.* The same argument with $z^\varepsilon \ell(z) \rightarrow \infty$ shows its infimum on $[\varepsilon, \infty)$ is positive.

The **key fact** is that a slowly varying function is eventually between $z^{-\varepsilon}$ and z^ε ; this follows from the representation theorem.

Potter v2 (also in Ex. 6) is proved the same way, applied to the ratio $\ell(x/y)/\ell(x)$. Use the representation $\ell(z) = c(z) \exp \int_{z_0}^z \frac{\epsilon(u)}{u} du$ with $\epsilon \rightarrow 0$: for large arguments,

$$\left| \log \frac{\ell(x/y)}{\ell(x)} \right| \leq \log \frac{c_{\max}}{c_{\min}} + \varepsilon |\log y|,$$

which gives the factor $y^{\pm\varepsilon}$. A compact-range argument then handles small arguments.

Slowly varying zoo.

- Slowly varying: constants, $\log z$, $\log \log z$, $1/\log z$, $e^{(\log z)^\eta}$ with $\eta < 1$, $1 + A \cos(\log z)/\log z$.
- *Not* slowly varying: z^ε , $e^{\sqrt{z}}$, $2 + \sin(\log z)$, $2 + \sin z$.

Slow variation does **not** require a limit to exist (e.g. $\log z$), but having a positive finite limit is sufficient.

6 §3.3 Statistics: Hill, tail processes and Weissman

6.1 Setup you must know cold

- Use an **intermediate sequence**: $k = k_n \rightarrow \infty$ (enough data) and $k/n \rightarrow 0$ (stay in the tail). Set $u_n = F^{\leftarrow}(1 - k/n)$, so that $n\overline{F}(u_n) = k$.

- The processes, from “ideal but unobservable” to “observable”:

$$\underbrace{\widehat{T}_n(s) = \frac{1}{k} \sum \mathbf{1}(Z_i > u_n s)}_{\text{deterministic level } u_n} \longrightarrow \underbrace{\widehat{T}_n(s) = \frac{1}{k} \sum \mathbf{1}(Z_i > Z_{n,n-k} s)}_{\text{random level, observable}}.$$

Both target $T(s) = s^{-1/\gamma}$, and $T_n(s) = \bar{F}(u_n s)/\bar{F}(u_n)$ is the mean of $\widetilde{T}_n$.

- The limits of the rescaled processes:

- $\sqrt{k}(\widetilde{T}_n - T_n) \xrightarrow{d} \mathbb{W} \circ T$, a time-changed Brownian motion;
- $\sqrt{k}(\widehat{T}_n - T) \xrightarrow{d} \mathbb{B} \circ T$, a time-changed Brownian bridge (the random level “pins” the process at $s = 1$);
- $\sqrt{k}(Z_{n,n-k}/u_n - 1) \xrightarrow{d} N(0, \gamma^2)$.

Why intermediate k ? (Example 3.3.) With a *fixed* threshold, $T_n(s)$ does not converge to $s^{-1/\gamma}$; it stays polluted by $\ell(u_n s)/\ell(u_n)$. You need $u_n \rightarrow \infty$ to kill ℓ .

6.2 The Hill estimator

$$\gamma_{n,k} = \frac{1}{k} \sum_{i=1}^k \log \frac{Z_{n,n-i+1}}{Z_{n,n-k}} = \int_1^\infty \frac{\widehat{T}_n(t)}{t} dt.$$

Motivation (three facts):

- (a) RV tails look Pareto beyond a high threshold;
- (b) if Y is Pareto($1/\gamma$) then $\log Y \sim \text{Exp}$ with mean γ ;
- (c) a mean is estimated by an average.

In fact, for an exact Pareto sample, $\gamma_{n,k}$ is the *MLE* of γ based on the top k excesses.

	Assumptions	Proof recipe
Consistency (3.13)	RV, intermediate k	Split $\int_{Z_{n,n-k}/u_n}^1 + \int_1^m + \int_m^\infty$. The first piece is $o_P(1)$ by T8, the second converges by T7, and the third is controlled by T10 with Potter and Markov.
CLT (3.14)	also $\sqrt{k}B_n(s_0) \rightarrow 0$ and the bias condition (3.7): $\sqrt{k} \int_1^\infty \frac{T_n - T}{t} dt \rightarrow 0$	Use the same three pieces. The near-1 piece is handled by the δ -method (T11) and gives $-\gamma \mathbb{W}(1)$. The bias piece vanishes by (3.7). The main piece is $\int \mathbb{W}(T)/t$ via T10 with Chebyshev. The variance γ^2 comes from T12.

Practical consequence: an approximate 95% CI is $\gamma_{n,k} \pm 1.96 \gamma_{n,k}/\sqrt{k}$.

Numerical example: Hill and Weissman by hand

$n = 1000$, $k = 5$. The top six order statistics are 50, 30, 20, 15, 12 and $Z_{n,n-5} = 10$.

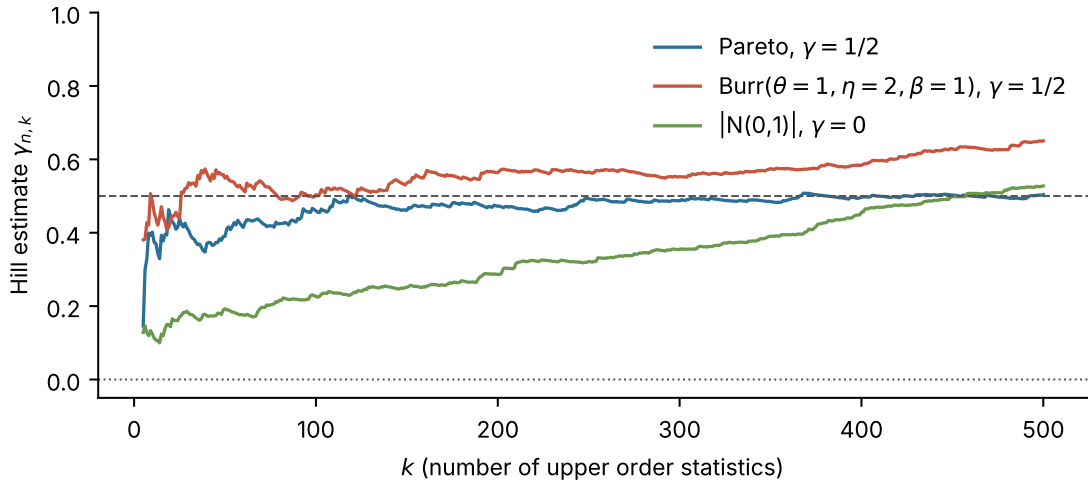
$$\begin{aligned} \gamma_{n,5} &= \frac{1}{5} (\log 5 + \log 3 + \log 2 + \log 1.5 + \log 1.2) \\ &= \frac{1}{5} (1.609 + 1.099 + 0.693 + 0.405 + 0.182) = \frac{3.989}{5} = 0.798. \end{aligned}$$

- **CI:** $0.798 \pm 1.96 \cdot 0.798/\sqrt{5} = [0.10, 1.50]$. This is useless, because $k = 5$ is far too small; that is the variance side of the trade-off.

- **Weissman quantile**, $p = 0.999$: $c = (1 - p)n/k = 0.2$, so $\widehat{Q}(0.999) = 0.2^{-0.798} \cdot 10 = 36.1$.
- **Weissman tail probability** at $z = 100$: $\widehat{F}(100) = \frac{5}{1000} \left(\frac{100}{10}\right)^{-1/0.798} = 2.8 \times 10^{-4}$.

6.3 Choosing k : the bias–variance trade-off and the Hill plot

- Small k gives high variance γ^2/k .
- Large k gives bias, because you include non-Pareto observations (the term $\int (T_n - T)/t$).
- **Hill plot**: plot $k \mapsto \gamma_{n,k}$, look for a stable (flat) region, and pick the largest k in it.



Simulated Hill plots, $n = 1000$. Pareto is flat because the bias is exactly zero. Burr drifts upward with k (second-order bias). For the half-normal ($\gamma = 0$) the Hill estimator is positive and drifting, so a Hill plot alone is not proof of a heavy tail.

T13. Verifying the bias condition (3.7) and finding admissible k (Prop. 3.1)

Recipe.

1. Find a **second-order expansion**: $\sup_{t \geq 1} t^{\beta^*} |\overline{F}(t) - ct^{-1/\gamma}| < \infty$ for some $\beta^* > 1/\gamma$. (The notes call it β .)
2. Deduce $u_n \asymp (n/k)^\gamma$ from $n\overline{F}(u_n) = k$.
3. Bound $|T_n(t) - t^{-1/\gamma}| \leq C \frac{n}{k} u_n^{-\beta^*} t^{-1/\gamma}$ for $t \geq 1$.
4. Integrate against dt/t and multiply by $\sqrt{k}$ to get $\sqrt{k} \cdot \frac{n}{k} (n/k)^{-\gamma\beta^*} = k^{(2\beta^*\gamma-1)/2} n^{1-\beta^*\gamma}$.
5. This tends to 0 iff

$$k = o(n^\rho), \quad \rho = \frac{2(\beta^*\gamma - 1)}{2\beta^*\gamma - 1}.$$

When to use it

Any question of the form “show (3.7) holds” or “find a suitable k ”: Prop. 3.1, Ex. 10, and in the conditional case Ex. 14–17.

Examples of the recipe

- **Exact Pareto**: $T_n \equiv T$, so there is no bias and any intermediate k works.
- **Burr** ($\theta = 1$): $\overline{F}(t) = t^{-\eta b}(1 + t^{-\eta})^{-b} = t^{-\eta b}(1 - bt^{-\eta} + O(t^{-2\eta}))$, writing b for the Burr

β . So $\beta^* = \eta b + \eta$, $\gamma = 1/(\eta b)$ and $\beta^* \gamma = 1 + 1/b$. This gives

$$\rho = \frac{2/b}{1 + 2/b} = \frac{2}{b + 2}.$$

For $b = 1, \eta = 2$ (Practical Ex. 1), $\rho = 2/3$, so take e.g. $k \approx n^{0.6}$.

- **Student t_ν :** the expansion has a relative error of order t^{-2} , so $\beta^* = \nu + 2$, $\beta^* \gamma = 1 + 2/\nu$ and $\rho = 4/(\nu + 4)$.
- **Log-perturbed tail (Example 3.4 / Ex. 10):** $\bar{F}(z) = z^{-1/\gamma} \log z$, so $T_n(t) = t^{-1/\gamma} \frac{\log(u_n t)}{\log u_n}$. Then

$$\sqrt{k} \int_1^\infty \frac{T_n - T}{t} dt = \frac{\sqrt{k} \gamma^2}{\log u_n} \sim \frac{\gamma \sqrt{k}}{\log(n/k)}.$$

The condition (3.7) therefore forces $k = o(\log^2 n)$: k cannot be a power of n . In Ex. 10 the remainder $O(t^{-\beta})$ adds a power-type term, which is negligible for such small k .

6.4 High quantiles and small probabilities (Weissman)

Two-step idea.

1. Estimate a moderately high quantile empirically: $Z_{n,n-k} \approx Q(1 - k/n)$.
2. Extrapolate with the Pareto tail using $\gamma_{n,k}$.

This gives

$$\hat{Q}(p) = Z_{n,n-k} \left(\frac{n(1-p)}{k} \right)^{-\gamma_{n,k}}, \quad \hat{F}(z) = \frac{k}{n} \left(\frac{z}{Z_{n,n-k}} \right)^{-1/\gamma_{n,k}}.$$

Asymptotics (Thms 3.15, 3.16). Let $c_n = n(1 - p_n)/k \rightarrow 0$; this is how far beyond the data you extrapolate. Then

$$\frac{\sqrt{k}}{\log(1/c_n)} \left(\frac{\hat{Q}(p_n)}{Q(p_n)} - 1 \right) \xrightarrow{d} N(0, \gamma^2).$$

The error of $\gamma_{n,k}$ gets *multiplied by the log-extrapolation distance*. Practical CI:

$$Q(p) \in \hat{Q}(p) \left(1 \pm 1.96 \gamma_{n,k} \log(1/c) / \sqrt{k} \right).$$

T14. Proof template for ratio estimators: $ABC - 1$

Write the ratio as a product $\hat{Q}/Q = A_n B_n C_n$, where

- A_n is the threshold error, $Z_{n,n-k}/u_n$;
- B_n is the index error, $c_n^{-(\gamma_{n,k} - \gamma)}$;
- C_n is the deterministic extrapolation error.

Then use the telescoping identity

$$ABC - 1 = (A - 1)BC + (B - 1)C + (C - 1).$$

Show that the first and third terms are negligible after scaling by $\sqrt{k}/\log(1/c_n)$: the first because it is only $O_P(1/\sqrt{k})$, the third by the rate condition (3.9). The middle term carries the CLT via the Taylor expansion $c_n^{-x} - 1 \approx x \log(1/c_n)$ and Slutsky.

When to use it

Any product-form estimator: Weissman quantile and probability estimators, and similar “extrapolation” estimators in exercises.

7 §3.4 Other estimators of γ

Estimator	Valid for	Asymptotic variance of $\sqrt{k}(\hat{\gamma} - \gamma)$	at $\gamma = \frac{1}{2}$
Hill	$\gamma > 0$	γ^2	0.25
Moment	$\gamma \in \mathbb{R}$	$1 + \gamma^2$ for $\gamma \geq 0$ (more complex for $\gamma < 0$)	1.25
GPD-MLE (POT)	$\gamma > -1/2$	$(1 + \gamma)^2$ (entry (1, 1) of $\Sigma(\gamma)$)	2.25
Pickands	$\gamma \in \mathbb{R}$	$\frac{\gamma^2(2^{2\gamma+1} + 1)}{(2(2^\gamma - 1) \log 2)^2}$	3.79

Key messages.

- If you *know* $\gamma > 0$, Hill has the smallest variance.
- Pickands uses only 3 order statistics, so it is very noisy.
- The moment estimator corrects Hill using $M_n^{(2)}$ and works for all γ .
- For $\gamma > 0$, $M_n^{(1)} \xrightarrow{\mathbb{P}} \gamma$ and $M_n^{(2)} \xrightarrow{\mathbb{P}} 2\gamma^2$. Then $1 - \frac{1}{2}(1 - \frac{\gamma^2}{2\gamma^2})^{-1} = 1 - 1 = 0$, so $\gamma_{n,k}^M \xrightarrow{\mathbb{P}} \gamma$. This is the punchline of Ex. 9 / Prop. 3.2.

Note on Pickands. The displayed denominator in the notes reads $(2(2^\gamma - 1) \log 2)^2$. The standard formula (de Haan & Ferreira, 2006) has 2^γ , as above; check with your lecturer.

Exercise 9 (moment estimator) in technique terms

1. Represent $M_n^{(2)} = 2 \int_1^\infty \frac{\log t}{t} \hat{T}_n(t) dt$ (T9).
2. Change level from $Z_{n,n-k}$ to u_n . This gives three pieces: a near-1 piece ($o_P(1)$ by T8), a main piece and a cross term $2 \log(Z_{n,n-k}/u_n) \cdot (\cdot) \xrightarrow{\mathbb{P}} 0$.
3. On $[1, m]$, uniform consistency gives $\rightarrow 2 \int_1^m \frac{\log t}{t} t^{-1/\gamma} dt \rightarrow 2\gamma^2$.
4. For the tail, use $\log t \leq t^\delta$ and Potter to get $\mathbb{E} \int_m^\infty \leq C m^{-1/\gamma+\delta+\varepsilon}$, then apply Markov (T10).

Exploratory tools (useful for Practical Ex. 1(c)):

- **Mean excess function** $e(u) = \mathbb{E}[Z - u \mid Z > u]$:
 - Pareto with $\gamma < 1$: $e(u) = \gamma u / (1 - \gamma)$, linear and increasing (asymptotically the same for any RV tail, by Karamata);
 - exponential: constant;
 - normal: decreasing, $\sim 1/u$;
 - lognormal: increasing, $\sim \sigma^2 u / \log u$;
 - Weibull with $a < 1$: increasing, $\sim u^{1-a}/a$.

The empirical version is volatile for large u (Thm 1.11).

- **Pareto QQ-plot:** plot $\log Z_{n,n-j+1}$ against $\log \frac{n+1}{j}$. An RV tail gives a straight line with slope $\approx \gamma$.

8 §3.5–3.6 Conditional extremes

Model. $\mathbb{P}(Y > y \mid X = u) = L_u(y) y^{-1/\gamma(u)}$ for u near x , so the tail index varies with the covariate.

Estimators. Use Nadaraya–Watson weights $w_j(x) = K(\frac{x-X_j}{h_n}) / \sum_i K(\frac{x-X_i}{h_n})$, which give $\bar{F}_n^x(y) = \sum_j w_j(x) \mathbf{1}(Y_j > y)$. Let $q_n^x = F_n^{\leftarrow, x}(1 - k_n/n)$ be the local empirical quantile. Then

$$\gamma_{n,k}(x) = \frac{n}{k_n} \sum_{j=1}^n w_j(x) \log_+ \frac{Y_j}{q_n^x} = \int_1^\infty \frac{\hat{T}_n^x(s)}{s} ds.$$

This is “Hill with kernel weights”, and the same identity $\gamma(x) = \int_1^\infty T^x(s)/s ds$ holds.

Main results.

- Consistency (Thm 3.21) under Condition 3.1.
- CLT (Thm 3.26):

$$\sqrt{k_n h_n} (\gamma_{n,k}(x) - \gamma(x)) \xrightarrow{d} N\left(0, \frac{\gamma(x)^2 \int K^2}{g(x)}\right).$$

- Effective sample size $k_n h_n$. A 95% CI is $\hat{\gamma}(x) \pm 1.96 \hat{\gamma}(x) \sqrt{\int K^2 / (g(x) k_n h_n)}$.
- For the *uniform kernel* on $[-1, 1]$ ($K = \frac{1}{2}$, $\int K^2 = \frac{1}{2}$) and a covariate transformed to $\text{Uniform}(0, 1)$ ($g = 1$), the variance is $\gamma(x)^2/2$. Be careful near the boundaries $x \approx 0, 1$, where the kernel window is truncated.

What each condition is for (Condition 3.1):

3.1.1 $1/\gamma(\cdot)$ Lipschitz	Neighbours within h_n have almost the same tail index.
3.1.2 $h_n \log u_n^x \rightarrow 0$	The exponent mismatch $(u_n)^{\pm ch_n} = e^{\pm ch_n \log u_n} \rightarrow 1$. A small change in γ is magnified by log of the threshold.
3.1.3 $\log L$ Lipschitz (relative to $\log y$)	The same for the slowly varying parts: $L_{x-uh}(y)/L_x(y) \in [y^{-ch}, y^{ch}]$.
3.1.4 $nh_n \bar{F}^x(u_n^x) = k_n h_n \rightarrow \infty$	The effective number of exceedances diverges, so the variance $\rightarrow 0$.

T15. Kernel substitution + Lipschitz sandwich + DCT

The template for Lemmas 3.1–3.2 and Ex. 11, 13:

$$\frac{\mathbb{E}[\mathbf{1}(Y > u_n^x t) K^q(\frac{x-X}{h_n})]}{h_n \bar{F}^x(u_n^x)} = \int K^q(u) \frac{\bar{F}^{x-uh_n}(u_n^x t)}{\bar{F}^x(u_n^x)} g(x - uh_n) du \quad (\text{substitute } z = x - uh_n).$$

Factor the ratio as

$$\underbrace{\frac{\bar{F}^x(u_n^x t)}{\bar{F}^x(u_n^x)}}_{\rightarrow t^{-1/\gamma(x)}} \cdot \underbrace{(u_n^x t)^{-(1/\gamma(x-uh)-1/\gamma(x))}}_{\in [e^{-ch \log(u_n t)}, e^{ch \log(u_n t)}] \rightarrow 1} \cdot \underbrace{\frac{L_{x-uh}(u_n^x t)}{L_x(u_n^x t)}}_{\rightarrow 1 \text{ by 3.1.3}}.$$

Also $g(x - uh_n) \rightarrow g(x)$, uniformly for $u \in [-1, 1]$ because K has compact support. Dominated convergence then gives $t^{-1/\gamma(x)} g(x) \int K^q$.

When to use it

Every expectation or variance computation in the conditional setting (Lemmas 3.1–3.3, Ex. 11, 13, 14(c)).

T16. Verifying Condition 3.2 and choosing (k_n, h_n) (Ex. 14–17)

Decompose the bias $\mathbb{E}\tilde{V}_n^x(t) - t^{-1/\gamma(x)}g(x)$ into three terms (Ex. 14(a)):

1. **Density smoothing:** $O(h_n)t^{-1/\gamma(x)}$. Taylor-expand g ; with a symmetric kernel even $O(h_n^2)$.
2. **Covariate shift of the tail:** $O(h_n \log(n/k_n))t^{-1/\gamma(x)+\varepsilon}$. This follows from T15 and $|e^z - 1| \leq |z| + \dots$
3. **Second-order tail bias:** $O(k_n^{\beta\gamma-1}n^{1-\beta\gamma})t^{-1/\gamma(x)}$. This is the unconditional recipe T13. Multiply by $\sqrt{k_n h_n}$; Condition 3.2 holds if each term vanishes. This gives (3.21):

$$\sqrt{k_n h_n^3} \log(n/k_n) \rightarrow 0, \quad \sqrt{h_n} k_n^{\beta\gamma-\frac{1}{2}} n^{1-\beta\gamma} \rightarrow 0.$$

Exercise 16: an explicit admissible choice

Try $k_n = n^a$, $h_n = n^{-b}$ with $0 < b < a < 1$ and write $\kappa = \beta_x \gamma(x) > 1$. The requirements are:

- $k_n h_n = n^{a-b} \rightarrow \infty$ (3.1.4);
- $h_n \log u_n \asymp n^{-b} \log n \rightarrow 0$ (3.1.2);
- $\sqrt{k_n h_n^3} \log n \rightarrow 0 \iff a < 3b$;
- $n^{-b/2+a(\kappa-\frac{1}{2})+1-\kappa} \rightarrow 0 \iff a < \frac{\kappa-1+b/2}{\kappa-1/2}$.

The last bound exceeds b exactly when $b < 1$. So for *any* $b \in (0, 1)$ you can pick $a \in (b, \min\{3b, \frac{\kappa-1+b/2}{\kappa-1/2}, 1\})$.

Concrete: $\kappa = 2$, $b = 0.3$ gives $a \in (0.3, 0.767)$, e.g. $k_n = n^{0.5}$, $h_n = n^{-0.3}$.

Exercise 17 (log-perturbed tail). The constraints become $\sqrt{k_n h_n^3} \log(n/k_n) \rightarrow 0$ and $\sqrt{k_n h_n} / \log(n/k_n) \rightarrow 0$, so $k_n h_n$ must grow slower than $\log^2 n$. A working choice is $k_n = (\log n)^3$, $h_n = (\log n)^{-2}$:

- $k_n h_n = \log n \rightarrow \infty$;
- $\sqrt{k_n h_n} / \log n = (\log n)^{-1/2} \rightarrow 0$;
- $\sqrt{k_n h_n^3} \log n = (\log n)^{-1/2} \rightarrow 0$;
- $h_n \log u_n \asymp (\log n)^{-1} \rightarrow 0$.

9 Exercise → technique map

Ex.	Task	Techniques / results
1	$\text{RV} \subset \mathcal{S} \subset \mathcal{L} \subset \text{heavy}$	event splitting + T1; conditioning; T4 (§3)
2	counterexamples	T3 (lognormal), lattice/jump construction, literature
3	$\gamma = \int T/s$; sum $\leftrightarrow$ integral Hill	T9
4	Weibull: heavy iff $a < 1$; not RV	T4, T3
5	Burr is RV, $\gamma = 1/(\eta\beta)$	T1
6	Potter bounds	SV properties, compact + tail split, representation thm
7	Breiman's lemma	T5 (Potter v2 + DCT), ratio telescoping
8	Var of Hill limit = γ^2	T12
9	moment estimator consistent	T9, T8, T7, T10
10	$k = o(\log^2 n)$	T13 (Example 3.4 calculation)
11	Lemma 3.2	T15
12	Var of conditional Hill limit	T12
13	limit in Thm 3.26 proof	T15 + Potter + Lipschitz (3.1.1, 3.1.3)
14–15	Condition 3.2 from second-order expansion	T16 (three-term bias), T13, T15
16–17	existence of (k_n, h_n)	T16 + power/log sequence choice
P1	Hill plot, ME plot, Weissman vs. empirical quantiles	§6, §7; ρ from T13
P2	Hill on fire claims; bootstrap vs. $N(\gamma, \gamma^2/k)$	Thm 3.14
P3	conditional Hill + bands	§8, CI with $\int K^2/(g k_n h_n)$

Generic proof skeleton for “show that estimator $\hat{\theta}_n$, a functional of tail processes, is consistent or asymptotically normal”:

1. Write $\hat{\theta}_n = \Phi(\hat{T}_n)$ as an integral (T9).
2. Switch the random level to the deterministic level: $\hat{T}_n(s) = \tilde{T}_n(sZ_{n,n-k}/u_n)$ (T8, T6).
3. Split the integral into the near-1 piece, $[1, m]$ and $[m, \infty)$.
4. Near 1: $Z_{n,n-k}/u_n \rightarrow 1$ with the δ -method (T11).
5. On $[1, m]$: (functional) LLN/CLT for $\tilde{T}_n$ (T7, bracketing CLT) and continuous mapping.
6. On $[m, \infty)$: Potter + Markov/Chebyshev, then Thm 1.5 (T10).
7. Bias: the deterministic piece vanishes by assumption, e.g. (3.7), which you verify via T13.
8. Combine with Slutsky and compute the variance (T12).

10 Classic pitfalls

Pitfalls

- **γ vs. $1/\gamma$.** $\bar{F}(z) = z^{-1/\gamma}\ell(z)$, so the Pareto exponent is $\alpha = 1/\gamma$. Burr: $\gamma = 1/(\eta\beta)$, not $\eta\beta$. Student t_ν : $\gamma = 1/\nu$.
- **Heavy-tailed $\neq$ RV.** Lognormal and Weibull($a < 1$) are heavy-tailed but in the *Gumbel* domain. Hill is not designed for them; it returns misleading positive values.
- **Hill needs $\gamma > 0$.** For $\gamma \leq 0$ use the moment, Pickands or GPD estimators.
- **Slowly varying $\neq$ convergent.** $\log z$ is slowly varying with no finite limit. Conversely, $2 + \sin(\log z)$ is bounded but *not* slowly varying.
- **Uniform convergence needs $t \geq a > 0$.** Near $t = 0$ the ratio $\bar{F}(tz)/\bar{F}(z)$ blows up, which is why processes live on $[s_0, \infty)$.
- **Intermediate k .** A fixed threshold gives a biased limit (Example 3.3); $k \sim n$ is not “in the tail”.
- **The bias condition can be brutal.** With $\ell = \log z$ you need $k = o(\log^2 n)$ (Example 3.4); with $\ell = e^{(\log z)^\eta}$ it is practically impossible.
- **Two sources of randomness in Hill.** The random level $Z_{n,n-k}$ contributes $-\gamma\mathbb{W}(1)$. Forgetting it gives the wrong variance ($2\gamma^2$ instead of γ^2).
- **Weissman rate.** The rate is $\sqrt{k}/\log(1/c_n)$, not $\sqrt{k}$: extrapolating further costs precision logarithmically.
- **Conditional rate.** The rate is $\sqrt{k_n h_n}$ and the variance has $\int K^2/g(x)$. In the conditional Hill estimator, k_n is a *global* level, and for the uniform kernel the local number of exceedances is $\approx 2k_n h_n g(x)$.

References used: M. Bladt, *Statistics for Insurance*, Ch. 3 and Ch. 1 (Thms 1.5, 1.10, 1.11, 1.14, 1.15, 1.17); Embrechts, Klüppelberg & Mikosch (1997), *Modelling Extremal Events*; de Haan & Ferreira (2006), *Extreme Value Theory*; Foss, Korshunov & Zachary (2013), *An Introduction to Heavy-Tailed and Subexponential Distributions*.